



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to S.R. ANANDARAMAN SON

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

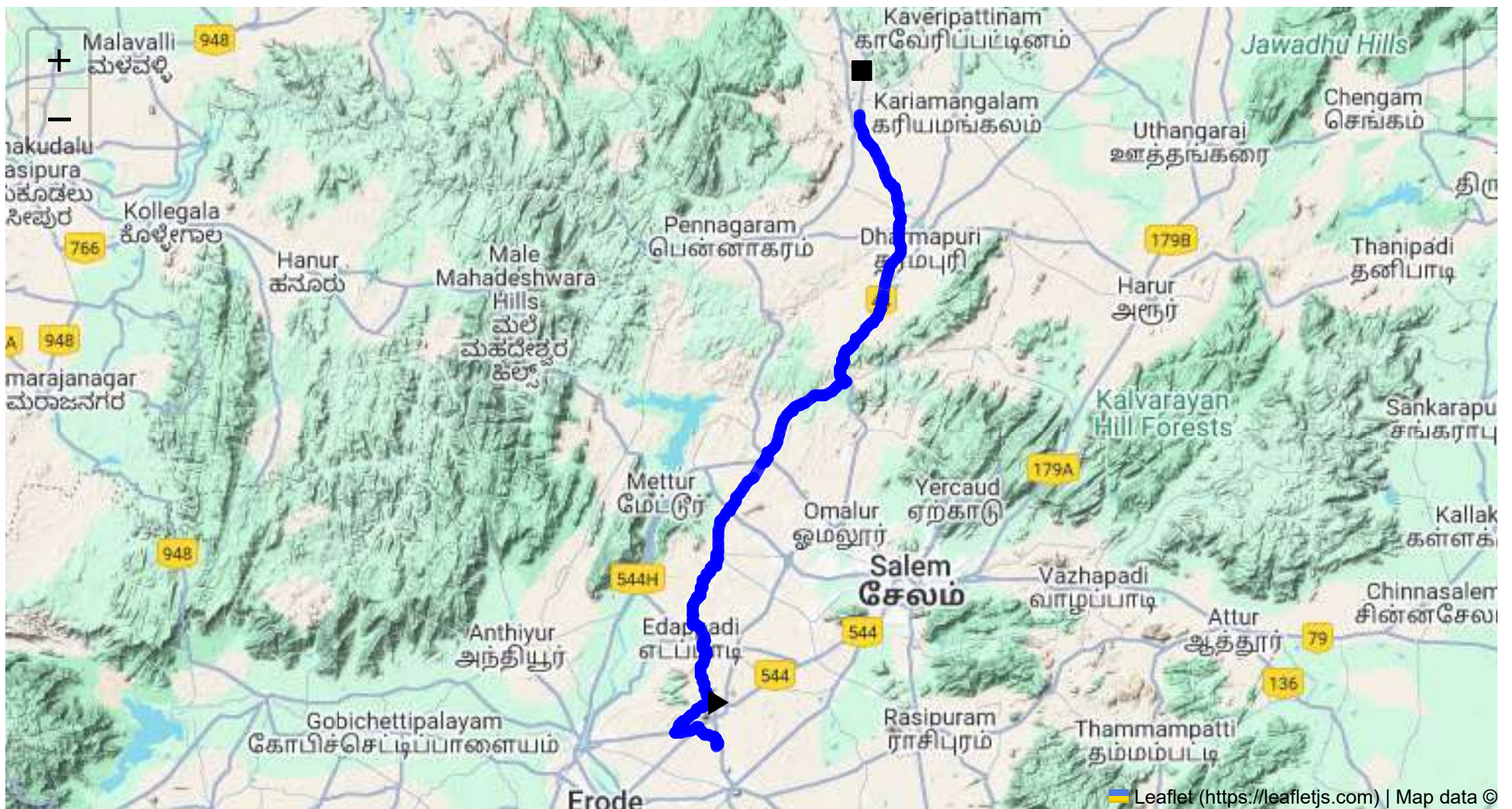
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 124.47 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (12.301595, 78.073721)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

This route begins at CVQF+23W in Sangagiri, Tamil Nadu, and concludes at Plot no 5/164, Theerthagiri Nagar, Karagathahalli, Palacode, Tamil Nadu. Major waypoints include Sankari RS - Erode Privu Road, Bhavani Main Road in Sankari West and Padaiveedu, Kurumbapatti, and Avanurkottai. It spans approximately 124.47 kilometers through a combination of local roads and state highways.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences a tropical climate. Weather hazards include:

- **Monsoon Rains (June to September):** Can cause road flooding and reduced visibility.
- **High Temperature (March to June):** May affect vehicle performance and driver fatigue.
- **Occasional Cyclones:** While rare, severe weather can lead to road closures.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM.
- **Congestion-Prone Areas:**
 - Sangagiri town center.
 - Sankari RS junction, due to the intersecting roads.
 - Padaiveedu, where local market traffic can cause delays.

4. Assessment of Road Quality and Infrastructure

- **Sangagiri to Sankari RS:** Generally well-maintained but narrow in some spots.
- **Bhavani Main Road (Sankari West to Padaiveedu):** Varies from good to moderate conditions, with some patchwork repairs.
- **Kurumbapatti to Avanurkottai:** Mostly rural roads, with potential for potholes and uneven surfaces.

5. Suggestions for Alternative Routes for Emergencies

In case of road blockages or extreme traffic, alternative routes include:

- Utilize state highways bypassing heavily congested towns.
- Local roads through nearby villages, accounting for slower travel but possible avoidance of major delays.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has stringent regulations regarding the transport of hazardous materials, including:

- Permits are required for transportation.
- Strict adherence to safety measures and proper labeling.
- Avoidance of densely populated and environmentally sensitive areas where possible.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been occasional incidents involving heavy vehicles in Tamil Nadu, primarily associated with:

- **Poor weather:** Often leading to skidding or overturning.
- **Road infrastructure failures:** Ineffective signage and lighting have contributed to accidents.

8. Environmental Considerations and Sensitive Areas

- **Near Water Bodies:** Caution around Bhavani River to avoid contamination.
- **Agricultural Zones:** Minimizing disturbances due to noise and emissions is crucial near farmland.

9. Analysis of Communication Coverage

- **Urban Areas (Sangagiri, Sankari):** Generally good mobile service.
- **Rural Segments (Kurumbapatti to Avanurkottai):** Potential for dead zones due to distance from cell towers.

10. Estimated Emergency Response Times

- **Urban Areas:** 15 to 30 minutes.
- **Rural Areas:** Could extend to 45 to 60 minutes or longer due to road access and distance.

11. Local Regulations

Local regulations pertinent to this route include permitted transport hours, speed limits specific to heavy vehicles, and enforcement patrols focused on load compliance.

12. An Overall Summary of Risk Assessment

The route presents moderate risk mainly due to varied road quality and potential traffic congestion. Weather conditions pose occasional hazards, particularly during monsoons. Emergency services are readily accessible in urban segments but less so in rural stretches. Proper preparedness for vehicle maintenance, weather conditions, and emergency procedures is essential. Overall, increased vigilance and adherence to regulations significantly mitigate risk for this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60359, 77.83965	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45764, 77.85219	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	High	11.45571, 77.81702	15 KM/Hr
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.46916, 77.82431	30 KM/Hr
11	Turn	Medium	11.49385, 77.86709	30 KM/Hr
12	Turn	High	11.49404, 77.86724	15 KM/Hr
13	Turn	Medium	11.49513, 77.86656	30 KM/Hr
14	Turn	Medium	11.50866, 77.86053	30 KM/Hr
15	Turn	Medium	11.52587, 77.85524	30 KM/Hr
16	Turn	Medium	11.52791, 77.85581	30 KM/Hr
17	Turn	Medium	11.54750, 77.85273	30 KM/Hr
18	Turn	Medium	11.55685, 77.85813	30 KM/Hr
19	Turn	Medium	11.56411, 77.85969	30 KM/Hr
20	Turn	High	11.57903, 77.85441	15 KM/Hr
21	Turn	Medium	11.57910, 77.85555	30 KM/Hr
22	Turn	High	11.58022, 77.85744	15 KM/Hr
23	Turn	Medium	11.58709, 77.85824	30 KM/Hr
24	Blind Spot	Blind Spot	11.60437, 77.83983	10 KM/Hr
25	Turn	Medium	11.61007, 77.84163	30 KM/Hr
26	Turn	Medium	11.61835, 77.84026	30 KM/Hr
27	Turn	Medium	11.64049, 77.84596	30 KM/Hr
28	Turn	Medium	11.64166, 77.84871	30 KM/Hr
29	Turn	High	11.67319, 77.86041	15 KM/Hr
30	Turn	Medium	11.67365, 77.86215	30 KM/Hr
31	Turn	High	11.68487, 77.86888	15 KM/Hr
32	Turn	High	11.68478, 77.87285	15 KM/Hr
33	Turn	High	11.69463, 77.87589	15 KM/Hr
34	Turn	High	11.69549, 77.87444	15 KM/Hr
35	Turn	High	11.76020, 77.89071	15 KM/Hr
36	Turn	High	11.75960, 77.89221	15 KM/Hr
37	Turn	Medium	11.83330, 77.94432	30 KM/Hr
38	Turn	Medium	11.83370, 77.94485	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
39	Turn	Medium	11.84179, 77.94732	30 KM/Hr
40	Turn	Medium	11.89747, 77.98079	30 KM/Hr
41	Turn	Medium	11.91901, 78.02623	30 KM/Hr
42	Turn	High	11.93618, 78.05392	15 KM/Hr
43	Turn	Medium	11.93710, 78.05390	30 KM/Hr
44	Turn	High	11.95824, 78.04981	15 KM/Hr
45	Turn	Medium	11.95992, 78.04947	30 KM/Hr
46	Turn	Medium	11.96320, 78.05139	30 KM/Hr
47	Turn	High	11.96401, 78.05134	15 KM/Hr
48	Turn	Medium	11.96504, 78.04975	30 KM/Hr
49	Turn	High	11.96597, 78.04920	15 KM/Hr
50	Turn	Medium	11.96679, 78.04942	30 KM/Hr
51	Turn	Medium	11.96804, 78.05080	30 KM/Hr
52	Turn	Medium	12.11993, 78.13279	30 KM/Hr
53	Turn	Medium	12.26532, 78.08427	30 KM/Hr
54	Turn	Medium	12.27200, 78.08089	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
10	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
13	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
14	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
15	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium
16	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60359, 77.83965



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49385, 77.86709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49404, 77.86724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49513, 77.86656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50866, 77.86053



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52587, 77.85524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52791, 77.85581



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54750, 77.85273



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55685, 77.85813



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56411, 77.85969



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57910, 77.85555



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58709, 77.85824



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.60437, 77.83983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61007, 77.84163



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61835, 77.84026



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64049, 77.84596



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64166, 77.84871



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67319, 77.86041



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67365, 77.86215



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68487, 77.86888



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68478, 77.87285



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69463, 77.87589



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69549, 77.87444



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.76020, 77.89071



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75960, 77.89221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83330, 77.94432



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83370, 77.94485



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.84179, 77.94732



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.89747, 77.98079



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91901, 78.02623



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.93618, 78.05392



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93710, 78.05390



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95824, 78.04981



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95992, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96597, 78.04920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96679, 78.04942



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.11993, 78.13279



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26532, 78.08427



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.27200, 78.08089
